

The Language of Medicine

13th edition

Davi-Allen Chabner

Chapter 3

Suffixes

Chapter Goals

- Define new suffixes and review those presented in previous chapters.
- Gain practice in word analysis by using these suffixes with combining forms to build and understand terms.
- Identify the functions of the different types of blood cells in the body.

Chapter 3

Lesson 3.1

Combining Forms (Slide 1 of 7)

Combining Form	Meaning
abdomin/o	abdomen
acr/o	extremities, top, extreme point
acu/o	sharp, severe, sudden
aden/o	gland
adip/o	fat
amni/o	amnion
angi/o	vessel
arteri/o	artery
arthr/o	joint
axill/o	armpit

Combining Forms (Slide 2 of 7)

Combining Form	Meaning
bi/o	life
blephar/o	eyelid
bronch/o	bronchial tubes
carcin/o	cancer
cardi/o	heart
chem/o	drug, chemical
chondr/o	cartilage
chron/o	time
col/o	colon (large intestine)
cyst/o	urinary bladder

Combining Forms (Slide 3 of 7)

Combining Form	Meaning
encephal/o	brain
erythr/o	red
hem/o	blood
hepat/o	liver
hydr/o	water, fluid
inguin/o	groin
isch/o	to hold back
lapar/o	abdomen, abdominal wall
laryng/o	larynx
leuk/o	white

Combining Forms (Slide 4 of 7)

Combining Form	Meaning
lymph/o	lymph
mamm/o	breast
mast/o	breast
morph/o	shape, form
muc/o	mucus
my/o	muscle
myel/o	spinal cord, bone marrow
necr/o	death (of cells or whole body)
nephr/o	kidney
neur/o	nerve

Combining Forms (Slide 5 of 7)

Combining Form	Meaning
neutr/o	neutrophil
nucle/o	nucleus
ophthalm/o	eye
oste/o	bone
ot/o	ear
path/o	disease
peritone/o	peritoneum
phag/o	to eat, swallow
phleb/o	vein
plas/o	formation, development

Combining Forms (Slide 6 of 7)

Combining Form	Meaning
pleur/o	pleura
pneumon/o	lungs
pulmon/o	lungs
radi/o	x-rays
rect/o	rectum
ren/o	kidney
rhin/o	nose
sarc/o	flesh
splen/o	spleen
staphyl/o	clusters

Combining Forms (Slide 7 of 7)

Combining Form	Meaning
strept/o	twisted chains
thorac/o	chest
thromb/o	clot
tonsill/o	tonsils
trache/o	trachea (windpipe)
ven/o	vein

Suffixes and Terminology – Noun Suffixes (Slide 1 of 7)

Suffix	Meaning
-algia	pain
-cele	hernia
-centesis	puncture to remove fluid
-coccus (plural: -cocci)	berry-shaped bacterium
-cyte	cell

Suffixes and Terminology – Noun Suffixes (Slide 2 of 7)

Suffix	Meaning
-dynia	pain
-ectomy	excision, removal, resection
-emia	blood condition
-genesis	condition of producing, forming
-gram	record

Suffixes and Terminology – Noun Suffixes (Slide 3 of 7)

Suffix	Meaning
-graph	instrument for recording
-graphy	process of recording
-itis	inflammation
-logy	study of
-lysis	breakdown, destruction, separation

Suffixes and Terminology – Noun Suffixes (Slide 4 of 7)

Suffix	Meaning
-malacia	softening
-megaly	enlargement
-oma	tumor, mass, collection of fluid
-opsy	to view
-osis	condition, usually abnormal

Suffixes and Terminology – Noun Suffixes (Slide 5 of 7)

Suffix	Meaning
-pathy	disease condition
-penia	deficiency
-phobia	fear
-plasia	development, formation, growth
-plasty	surgical repair

Suffixes and Terminology – Noun Suffixes (Slide 6 of 7)

Suffix	Meaning
-ptosis	drooping, falling, prolapse
-rrhea	flow, discharge
-sclerosis	hardening
-scope	instrument for visual examination
-scopy	process of visual examination (with an endoscope)
-stasis	controlling, stopping

Suffixes and Terminology – Noun Suffixes (Slide 7 of 7)

Suffix	Meaning
-stomy	opening to form a mouth (stoma)
-therapy	treatment
-tomy	incision, cutting into
-trophy	development, nourishment

QUICK QUIZ (Slide 1 of 3)

1. Which term means fear of heights?

- A. Acromegaly
- B. Hydrophobia
- C. Acrophobia
- D. Agoraphobia

Suffixes and Terminology – Shorter Noun Suffixes

Suffix	Meaning
-er	one who
-ia	condition
-ist	specialist
-ole	little, small
-ule	little, small
-um, -ium	structure, tissue
-us	structure, substance
-y	condition, process

Suffixes and Terminology – Adjective Suffixes (Slide 1 of 2)

Suffix	Meaning
-ac, -iac	pertaining to
-al	pertaining to
-ar	pertaining to
-ar	pertaining to
-eal	pertaining to
-genic	pertaining to producing, produced by, or in

Suffixes and Terminology – Adjective Suffixes (Slide 2 of 2)

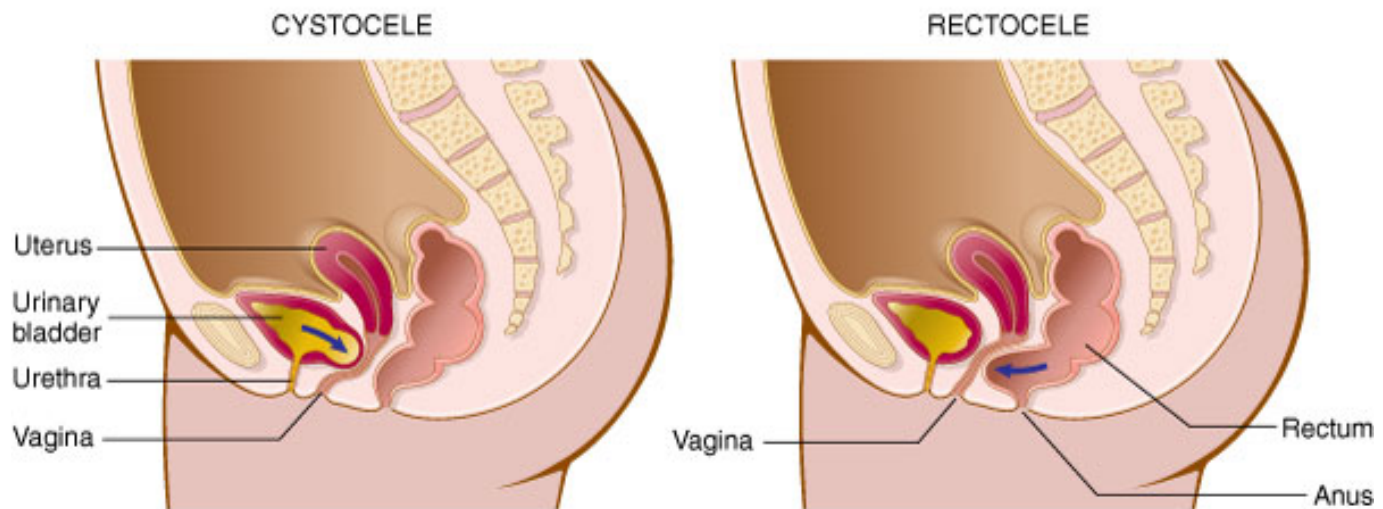
Suffix	Meaning
-ic, -ical	pertaining to
-oid	resembling
-ose	pertaining to, full of
-ous	pertaining to
-tic	pertaining to

Build Terms with Suffixes and Combining Forms

Combining Form + Suffix	Meaning
arthr/itis	inflammation of joint
blephar/o/ptosis	drooping eyelid
isch/emia	loss of normal blood flow
oste/o/genic	produced by bone tissue
staphyl/o/cocci	berry-shaped bacteria that grow in clusters

A Closer Look: Hernia

- Protrusion of an organ or the muscular wall of an organ through the cavity that normally contains it



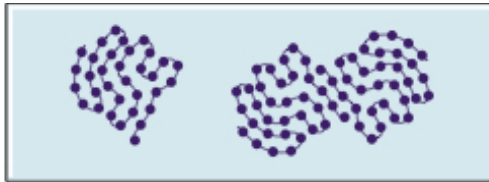
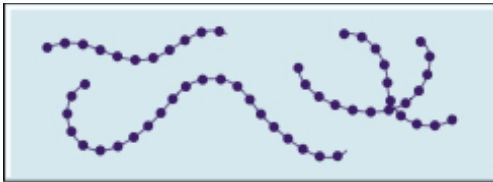
QUICK QUIZ (Slide 2 of 3)

2. A rectocele is:

- A. a herniation of the intestines
- B. protrusion of a portion of the rectum toward the vaginal wall
- C. a hiatal hernia
- D. an omphalocele

A Closer Look: Streptococcus

- **Streptococcus and Staphylococcus:**
Bacteria that grow in chains or clusters



A Closer Look: Blood Cells

(Slide 1 of 8)

1. Erythrocytes

- Red blood cells

2. Leukocytes

- White blood cells
- Clotting cells

3. Thrombocytes or platelets

A Closer Look: Blood Cells

(Slide 2 of 8)

1. Erythrocytes

- Red blood cells
- Made in bone marrow
- Carry oxygen from the lungs to all body cells
- Hemoglobin: blood protein

A Closer Look: Blood Cells (Slide 3 of 8)

2. Leukocytes

- White blood cells
- Five different kinds of leukocytes, divided into two categories
 - Granulocytes
 - Mononuclear leukocytes

A Closer Look: Blood Cells

(Slide 4 of 8)

Granulocytes – three types

- Eosinophils
- Basophils
- Neutrophils

A Closer Look: Blood Cells

(Slide 5 of 8)

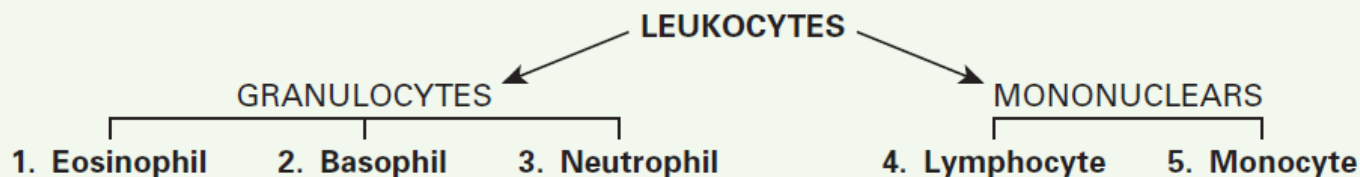
Mononuclear cells – two types

- Lymphocytes
- Monocytes

A Closer Look: Blood Cells

(Slide 6 of 8)

TABLE 3-1 FIVE TYPES OF LEUKOCYTES (WHITE BLOOD CELLS)



A Closer Look: Blood Cells (Slide 7 of 8)

3. Thrombocytes

- Also known as platelets
- Tiny fragments of blood cells that are formed in the bone marrow
- Necessary for blood clotting

A Closer Look: Blood Cells (Slide 8 of 8)

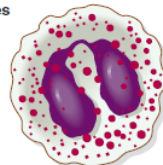
ERYTHROCYTES
(no nucleus;
contain hemoglobin)



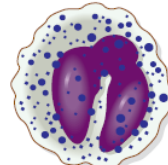
LEUKOCYTES

(polymorphonuclears)

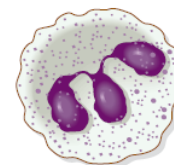
Granulocytes



1. Eosinophil
(granules stain red)



2. Basophil
(granules stain blue)

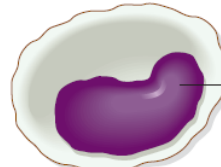


3. Neutrophil
(granules stain pale purple;
polymorphonuclear leukocyte,
or poly)

Mononuclear cells



4. Lymphocyte



5. Monocyte

One large nucleus

**THROMBOCYTES
(PLATELETS)**



QUICK QUIZ (Slide 3 of 3)

3. Which type of blood cells are necessary for clotting?

- A. Erythrocytes
- B. Granulocytes
- C. Monocytes
- D. Thrombocytes

A Closer Look: Acromegaly

- Endocrine disorder
- Occurs when the **pituitary gland** produces an excessive amount of growth hormone *after* completion of puberty
- Often results from a benign tumor of the pituitary gland
- Person typically is of normal height, but bones and soft tissue in the hands, feet, and face grow abnormally
- Abraham Lincoln was believed to have features of acromegaly

A Closer Look: Laparoscopy

- A form of **minimally invasive surgery**
- Visual examination of the abdominal cavity using a laparoscope, a lighted telescopic instrument
- Surgeon inserts the laparoscope through an incision in the abdomen near the navel
- Used to examine abdominal viscera for evidence of disease or for various procedures

A Closer Look: Adenoids

- Small masses of lymphatic tissue
- Located in the part of the pharynx near the nose and nasal passages
- The literal meaning, “resembling glands,” is appropriate since they are neither endocrine nor exocrine glands
- Adenoidectomy may be needed if they become enlarged and block the airway from the nose to the pharynx